Linear Equations & Systems

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

If $2(x-6)=3x-5$, what is the value of x?

Answer: -7. Distribute: $2x-12=3x-5$. $-x=-5+12 \Rightarrow x=-7$.

2 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

$\{x+6y=-43;\ -x-y=13$ What is the value of y?

Answer: -6. Adding the equations eliminates x: $5y=-30 \Rightarrow y=-6$. Then $x=-7$.

3 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

$\{6x-3y=-6;\ -6x+5y=-2$ What is the value of y?

Answer: -4. Adding the equations eliminates x: $2y=-8 \Rightarrow y=-4$. Then $x=-3$.

4 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

If $3(x-2)=-3x-12$, what is the value of x?

Answer: -1. Distribute: $3x-6=-3x-12$. $6x=-12+6 \Rightarrow x=-1$.

5 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

If $2x+5=-5x+47$, what is the value of x?

Answer: 6. Applying Linear Equations in One Variable & Systems of Equations principles, $7x=47-5 \Rightarrow x=6$ Therefore, the verified correct option is 6.

6 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

If $6(x+1)=-x+20$, what is the value of x?

Answer: 2. Distribute: $6x+6=-x+20$. $7x=20-6 \Rightarrow x=2$.

7 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

What is the sum of all solutions to $|3x-9|=15$?

Answer: 6. Two cases: $3x-9=15 \Rightarrow 3x=24 \Rightarrow x=8$; and $3x-9=-15 \Rightarrow 3x=-6 \Rightarrow x=-2$. The sum of solutions is $8+(-2)=6$.

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Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

If $2(x + 7) = 3x + 19$, what is the value of x?

Answer: -5. Distribute: $2x + 14 = 3x + 19$. $-x = 19 - 14 \Rightarrow x = -5$.

9 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

If $2(x - 5) = 4x - 12$, what is the value of x?

Answer: 1. Distribute: $2x - 10 = 4x - 12$. $-2x = -12 + 10 \Rightarrow x = 1$.

10 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

$\{2x - 2y = 8;\ -2x + 3y = -7$ What is the value of y?

Answer: 1. Adding the equations eliminates x: $y = 1 \Rightarrow y = 1$. Then $x = 5$.

11 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

If $2(x + 1) = 3x - 1$, what is the value of x?

Answer: 3. Distribute: $2x + 2 = 3x - 1$. $-x = -1 - 2 \Rightarrow x = 3$.

12 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

If $8x - 7 = -2x + 73$, what is the value of x?

Answer: 8. Applying Linear Equations in One Variable & Systems of Equations principles, $10x = 73 + 7 \Rightarrow x = 8$ Therefore, the verified correct option is 8.

13 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

$\{2x + 6y = -48;\ -2x + 3y = -33$ What is the value of x?

Answer: 3. Adding the equations eliminates x: $9y = -81 \Rightarrow y = -9$. Then $x = 3$.

14 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

$\{5x - 5y = -10;\ -5x - 4y = 64$ What is the value of $x + y$?

Answer: -14. Adding the equations eliminates x: $-9y = 54 \Rightarrow y = -6$. Then $x = -8$.

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

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15 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations
$\{2x - 5y = -9;\ -2x - 2y = 16$ What is the value of y?

Answer: -1. Adding the equations eliminates x: $-7y = 7 \Rightarrow y = -1$. Then $x = -7$.

16 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations
If $5x - 10 = 4x - 6$, what is the value of x?

Answer: 4. Applying Linear Equations in One Variable & Systems of Equations principles, $x = -6 + 10 \Rightarrow x = 4$ Therefore, the verified correct option is 4.

17 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations
$\{x + 5y = -27;\ -x - 2y = 9$ What is the value of $x + y$?

Answer: -3. Adding the equations eliminates x: $3y = -18 \Rightarrow y = -6$. Substituting gives $x = 3$. Therefore, $x + y = 3 + (-6) = -3$.

18 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations
$\{x - y = -1;\ -x + 3y = -1$ What is the value of x?

Answer: -2. Adding the equations eliminates x: $2y = -2 \Rightarrow y = -1$. Then $x = -2$.

19 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations
$\{x - 4y = -34;\ -x + 6y = 50$ What is the value of $x + y$?

Answer: 6. Adding the equations eliminates x: $2y = 16 \Rightarrow y = 8$. Then $x = -2$.

20 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations
Consider the system: $\{3x + 2y = 19; 2x - y = 8$ What is the value of $x + y$?

Answer: 7. From second equation, $y = 2x - 8$. Substitute into first:
$3x + 2(2x - 8) = 19 \Rightarrow 7x - 16 = 19 \Rightarrow 7x = 35 \Rightarrow x = 5$. Then $y = 2(5) - 8 = 2$. So $x + y = 5 + 2 = 7$.

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

21 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations
If $3x + 14 = 32$, what is the value of x?

a. 15.333333333333334

b. 18

c. 6 ✓

d. 7

Subtract 14 from both sides to get $3x = 18$, then divide by 3: $x = 6$.

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22 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $2(x - 8) = 8$, what is the value of x?

a. 0

b. 12 ✓

c. 4

d. -4

Divide both sides by 2: $x - 8 = 4$. Add 8: $x = 12$.

23 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

A moving company charges a one-time fee of USD 30 plus USD 55 per mile. If the total charge was USD 305, how many miles was the move?

a. 5 ✓

b. 6

c. 4

d. 7

Let n be the number of miles. Then $30 + 55n = 305$, so $55n = 275$ and $n = 5$.

24 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $\frac{x}{3} + 5 = 13$, what is the value of x?

a. 27

b. 24 ✓

c. 8

d. 39

Subtract 5: $\frac{x}{3} = 8$. Multiply by 3: $x = 24$.

25 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $8x + 12 = 4x + 36$, what is the value of x?

a. 6 ✓

b. 2

c. 24

d. 8

Subtract $4x$ and 12 from both sides: $4x = 24$, so $x = 6$.

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26 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $5x = 15$, what is the value of $10x$?

a. 30 ✓

b. 3

c. 15

d. 6

Multiply both sides of $5x = 15$ by 2: $10x = 30$.

27 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

A taxi company charges a one-time fee of USD 25 plus USD 30 per mile. If the total charge was USD 175, how many miles was the trip?

a. 6

b. 4

c. 7

d. 5 ✓

Let n be the number of miles. Then $25 + 30n = 175$, so $30n = 150$ and $n = 5$.

28 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $\frac{x}{2} + 3 = 9$, what is the value of x?

a. 12 ✓

b. 6

c. 18

d. 14

Subtract 3: $\frac{x}{2} = 6$. Multiply by 2: $x = 12$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $7x - 8 = 8x - 15$, what is the value of x?

a. 11

b. 3

c. 7 ✓

d. 13

Applying Linear Equations in One Variable & Systems of Equations principles, $-1x = -15 + 8 \Rightarrow x = 7$

Therefore, the verified correct option is 7.

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30 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

$\{5x + y = -49;\ -5x - 4y = 76$ What is the value of $x + y$?

a. -18

b. -17 ✓

c. -10

d. -9

Adding the equations eliminates x: $-3y = 27 \Rightarrow y = -9$. Then $x = -8$.

31 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $8x + 1 = -x - 26$, what is the value of x?

a. 4

b. 3

c. -3 ✓

d. -10

Applying Linear Equations in One Variable & Systems of Equations principles, $9x = -26 - 1 \Rightarrow x = -3$

Therefore, the verified correct option is -3.

32 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $8x + 2 = -4x - 58$, what is the value of x?

a. -5 ✓

b. -1

c. 4

d. -10

$12x = -58 - 2 \Rightarrow x = -5$

33 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $7x - 2 = -8x + 28$, what is the value of x?

a. 3

b. 1

c. -3

d. 2 ✓

Applying Linear Equations in One Variable & Systems of Equations principles, $15x = 28 + 2 \Rightarrow x = 2$

Therefore, the verified correct option is 2.

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34 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

$\{x + 5y = 24;\ -x + 2y = 18$ What is the value of $x + y$?

a. 7

b. 3

c. 0 ✓

d. -8

Adding the equations eliminates x: $7y = 42 \Rightarrow y = 6$. Then $x = -6$.

35 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

$\{6x - 6y = 12;\ -6x - 2y = 12$ What is the value of x?

a. -1 ✓

b. -3

c. -5

d. -6

Adding the equations eliminates x: $-8y = 24 \Rightarrow y = -3$. Then $x = -1$.

36 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $\frac{2}{x} + \frac{3}{2x} = \frac{7}{12}$, what is the value of x?

a. 4

b. 5

c. 6 ✓

d. 7

Common denominator is $2x$: $\frac{4+3}{2x} = \frac{7}{2x} = \frac{7}{12} \Rightarrow 2x = 12 \Rightarrow x = 6$.

37 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

A coffee shop sells small coffees for USD 2.50 and large coffees for USD 4.00. On Monday, the shop sold 80 total coffees for USD 260. How many large coffees were sold?

a. 35

b. 40 ✓

c. 45

d. 50

Let $s + l = 80$ and $2.5s + 4l = 260$. From first, $s = 80 - l$. Substitute:

$2.5(80 - l) + 4l = 260 \Rightarrow 200 + 1.5l = 260 \Rightarrow 1.5l = 60 \Rightarrow l = 40$.

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

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38 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

$\{4x + 5y = 13;\ -x + 6y = 4$ What is the value of x?

a. 9

b. 2 ✓

c. 0

d. 11

Solving the system by elimination gives $x = 2$ and $y = 1$.

39 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

$\{4x + 6y = 4; 12x + ky = -15$ For what value of k does this system of equations have no solution?

a. 20

b. 21

c. 18 ✓

d. 13

The lines are parallel (no solution) when the coefficients are proportional: $\frac{12}{4} = \frac{k}{6} \Rightarrow k = 18$ (and the constants are not in that same ratio).

40 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

$\{4x + 2y = 2; 8x + ky = -10$ For what value of k does this system of equations have no solution?

a. 9

b. 4 ✓

c. 10

d. 5

The lines are parallel (no solution) when the coefficients are proportional: $\frac{8}{4} = \frac{k}{2} \Rightarrow k = 4$ (and the constants are not in that same ratio).

41 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

$\{6x + 5y = 4;\ -2x - y = -4$ What is the value of x?

a. 4 ✓

b. 2

c. 9

d. 10

Solving the system by elimination gives $x = 4$ and $y = -4$.

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42 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

What is the sum of all solutions to the equation $|2x + 2| = 10$?

a. -5

b. -10

c. -2 ✓

d. -3

$2x + 2 = 10$ gives $x = 4$; $2x + 2 = -10$ gives $x = -6$. Sum $= -2$.

43 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

If $2x + 3y = 14$ and $5x + 4y = 28$, what is the value of $7x + 7y$?

a. 38

b. 42 ✓

c. 46

d. 52

Add the two equations directly: $(2x + 3y) + (5x + 4y) = 14 + 28 \Rightarrow 7x + 7y = 42$.

44 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

For what value of p does the equation $4(2x - 3) + p = 8x - 5$ have infinitely many solutions?

a. -7

b. -5

c. 5

d. 7 ✓

Expand left side: $8x - 12 + p = 8x - 5$. Subtract $8x$: $-12 + p = -5 \Rightarrow p = 7$.

45 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

A chemist needs to mix a 20% acid solution with a 50% acid solution to produce 60 liters of a 35% acid solution. How many liters of the 50% solution are required?

a. 25

b. 30 ✓

c. 35

d. 40

Let x be liters of 20% solution and y be liters of 50% solution. $x + y = 60$ and $0.20x + 0.50y = 0.35(60) = 21$. From first, $x = 60 - y$. Substitute:

$0.20(60 - y) + 0.50y = 21 \Rightarrow 12 + 0.30y = 21 \Rightarrow 0.30y = 9 \Rightarrow y = 30$.

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